

Course name: Data Communication LAB [K], Course code: COE 3201

Semester: Spring 2025-2026, Name of the Faculty: Dr. Muhammad Morshed Alam

Experiment no: 04

Experiment name: Study of Analog to Digital Conversion using MATLAB

Group no: 05

SL	Student name	ID	Marks
1	Maisha Mahjabin	23-54978-3	
2	Israt Jahan Surovy	23-54972-3	
3	Chowdhury Zarin Tasnim Oishy	23-54720-3	
4	Tanjimul Hossain Mojumder	23-52826-2	
5	Showmikul Islam Lemo	23-52830-2	

Marking Rubrics (to be filled by Faculty):

Category	Proficient [5]	Good [4]	Acceptable [3-2]	Unacceptable [1]	Secured Marks
Objective/ related theory	Explains all necessary and relevant theoretical background information, measures, and variables.	Explains the important parts of theoretical background information, may be partially irrelevant or need more details.	Explains some theoretical background information, but some information may be missing, irrelevant or inaccurate.	A lot of information is missing, out of context and/or inaccurate.	
Procedure/ code	- Working procedure or code is clearly presented and is supported by proper comments. - Methods are clearly written, including all steps in sufficient detail for the experiment to be repeated.	- Working procedure or code is given for the experiment to be sufficient. - Methods are correct but the steps may be lacking in detail, making the experiment hard to be repeated.	- Working procedure or code is missing some steps and/or contains some mistakes. - Simulation output has many problems.	- Working procedure or code is absent or missing major steps and/or contains mistakes. - Simulation output completely wrong.	
Experiment results/table/simulation output figures	- Clear, accurate diagrams and graphs are labeled neatly and accurately with excellent detail. - Simulated result meets all criteria; outcomes are described clearly and accurately.	- Diagrams and plots are included and are correctly labeled in brief, but there may be some lack of clarity. - Most criteria are met, but there may be some lack of clarity and/or incorrect information.	- Diagrams and plots are included and are labeled, minor mistakes may be present. - Results do not match exactly with the theoretical values and/or analysis is unclear.	- Needed plots/diagrams are missing or are missing important labels. - Experimental results are missing or completely incorrect.	

Data Interpretation/analysis/question answer/conclusion	• Interpretation and analysis of related outcomes (consequences and implications) are logical. • Any report questions are properly answered with detailed justification or calculations.	• Analysis is logically tied to information; some related outcomes are not clear. • Report questions are answered correctly but may be lacking detail or contain minor logical error.	• Analysis is inconsistently tied to some of the information discussed; related outcomes are oversimplified. • Report questions are answered but contain wrong information or major logical error.	• Only the data was reported, there is no analysis. • Report questions are missing or are completely wrong.	
Comments:				Total Marks (Out of 20):	

Title: Study of Analog to Digital Conversion using MATLAB.

Objectives:

The objective of this experiment is to study how analog signals can be converted into digital signals using MATLAB. The experiment demonstrates the processes of sampling, quantization, and encoding in Pulse Code Modulation (PCM) and helps understand the representation of analog signals in digital form.

Theory:

Analog to Digital Conversion:

Digital signals are better than analog signals. Today, the trend is to change an analog signal into digital form. In this section, we describe two methods: pulse code modulation and delta modulation. After we create digital data (digitization), we can use line coding to turn the digital data into a digital signal.

Pulse Code Modulation (PCM):

The most common way to change an analog signal into digital data (digitization) is called pulse code modulation (PCM). A PCM encoder works in three steps:

- The analog signal is sampled.
- The sampled signal is quantized.
- The quantized values are written as streams of bits.

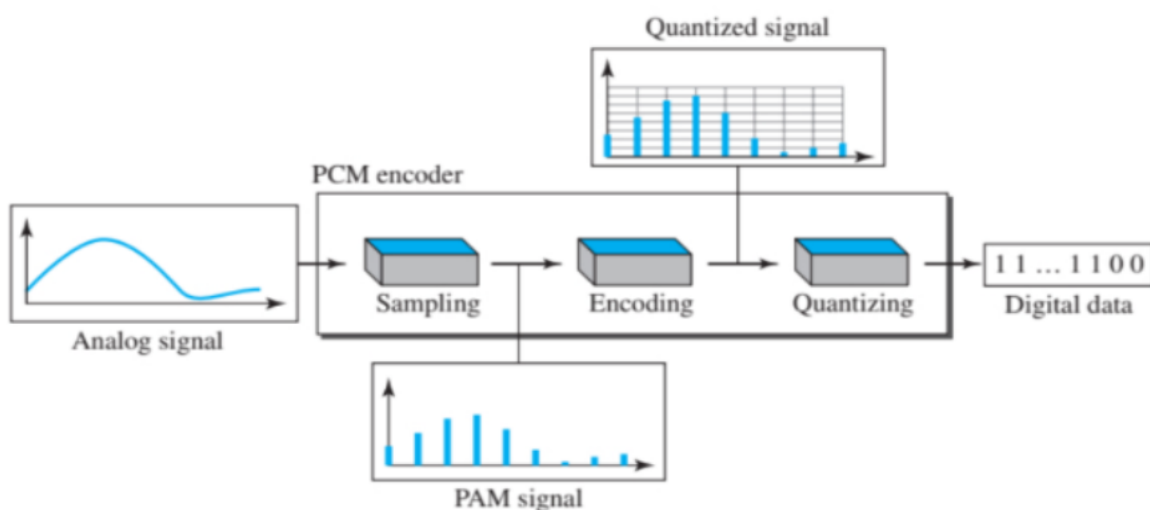


Fig: Components of PCM encoder

Sampling:

The first step in PCM is sampling. The analog signal is sampled every T_s seconds, where T_s is the sampling interval or period. The opposite of the sampling interval is called the sampling rate or frequency, shown as f_s , where $f_s = 1/T_s$. There are three types of sampling—ideal, natural, and flat-top.

Code:

23-54972-3 (AB-CDEFG-H)

$$a_1 = F+1 = 7+1=8, a_2 = F+3 = 7+3= 10, a_3 = F+2= 7+2=9, a_4 = F+4= 7+4=11,$$

$$f_1 = G+5=2+5=7, f_2 = G+7= 2+7= 9, f_3 = G+1= 2+1=3, f_4 = G+2= 2+2=4$$

$$\begin{aligned} \text{sig} &= a_1 \sin(2\pi f_1 t) + a_2 \cos(2\pi f_2 t) + a_3 \sin(2\pi f_3 t) + a_4 \sin(2\pi f_4 t) \\ &= 8\sin(2\pi \cdot 7t) + 10\cos(2\pi \cdot 9t) + 9\sin(2\pi \cdot 3t) + 11\sin(2\pi \cdot 4t) \end{aligned}$$

a)

```
clc;
clear;
time_duration= 0.1;
analog_t = 0:0.0001:time_duration;
analog_sig = 8*sin(2*pi*7*analog_t) + 10*cos(2*pi*9*analog_t) + 9*sin(2*pi*3*analog_t) + 11*sin(2*pi*4*analog_t);

% sampling
fs=150;
samp_t=0:1/fs:time_duration;
samp_sig = 8*sin(2*pi*7*samp_t) + 10*cos(2*pi*9*samp_t) + 9*sin(2*pi*3*samp_t) + 11*sin(2*pi*4*samp_t);

subplot(3,1,1)
plot(analog_t,analog_sig)
xlabel('Time(sec)')
ylabel('A(v)')
title('Input analog signal')

subplot(3,1,2)
stem(samp_t,samp_sig)
xlabel('Time(sec)')
ylabel('A(v)')
title('Sampled signal')

% quantization
nb=6;
L=2^nb-1;
delta= (max(samp_sig)-min(samp_sig))/L;
i=round((samp_sig-min(samp_sig))/delta);
xq=min(samp_sig)+i*delta;

subplot(3,1,3)
stairs(samp_t,xq)
xlabel('Time(sec)')
ylabel('A(v)')
title('Quantized signal')
```

b)

```
% Encoding ( analog to digital data)

dig_sig_matrix=de2bi(i,nb);
% encoded digital signal

dig_sig=reshape (dig_sig_matrix',1,[]);|
```

c)

Sampling Frequency:

Nyquist theorem:

$$f_s \geq 2f_{max}$$

Largest frequency:

$$f_{max} = G + 7 = 2 + 7 = 9$$

So,

$$f_s \geq 2f_{max}$$

$$f_s \geq 2(9) = f_s \geq 18$$

Number of Quantization Bits (Using SNR):

Given SNR = 25 dB

$$SNR(dB) = 6.02n + 1.76$$

$$25 = 6.02n + 1.76$$

$$n = \frac{25-1.76}{6.02}$$

$$n \approx 4 \text{ bits}$$

Quantization Levels:

$$L = 2^n = 2^4 = 16 \text{ levels}$$

Bit rate:

$$BR = f_s \times n_b$$

$$= 18 \times 4 = 72 \text{ bits/sec}$$

For PCM:

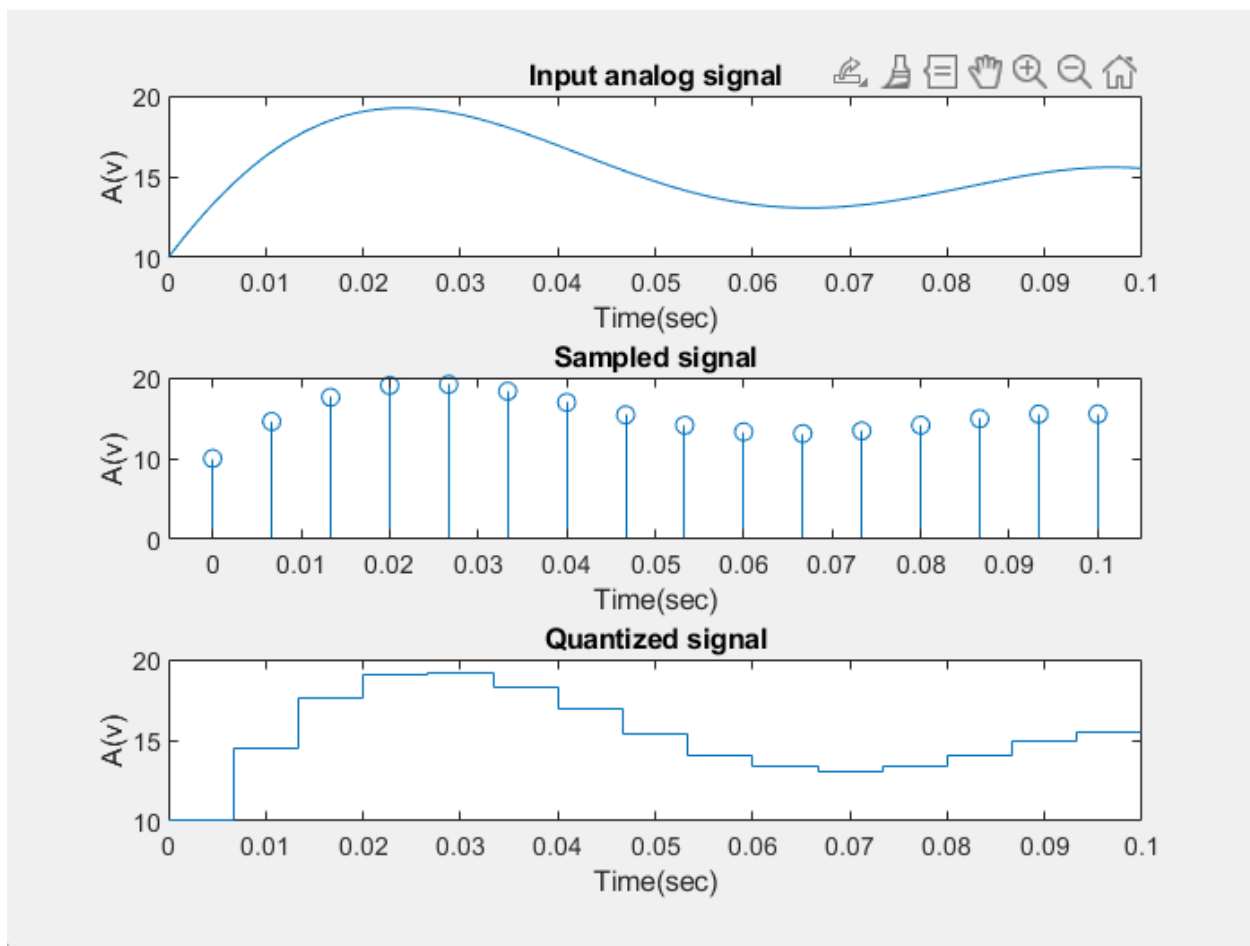
$$\text{Bandwidth} = \frac{\text{Bitrate}}{2} = \frac{72}{2} = 36 \text{ Hz}$$

Given channel bandwidth = 150 Hz

Required bandwidth = 36 Hz

Simulation figure and table:

a)



b)

```
dig_sig =  
  
Columns 1 through 17  
0    0    0    0    0    0    1    1    1    1    1    0    0    0    1    0    1  
  
Columns 18 through 34  
1    0    1    1    1    1    1    1    1    1    1    1    1    1    0    0    1  
  
Columns 35 through 51  
1    1    0    0    0    0    1    1    1    0    1    0    0    1    0    0    1  
  
Columns 52 through 68  
1    1    0    1    1    1    0    1    0    1    0    1    0    1    0    1    1  
  
Columns 69 through 85  
1    0    1    0    0    0    1    1    1    0    0    1    0    0    0    1    0  
  
Columns 86 through 96  
1    1    0    0    1    0    1    1    0    0    1  
  
fx >>
```

Result analysis/discussion:

From the result, the analog-to-digital conversion process was clearly observed using Pulse Code Modulation (PCM). The analog signal appeared as a smooth continuous waveform, and after sampling, discrete points followed the original signal shape closely when the sampling frequency satisfied the Nyquist condition. This confirmed that proper sampling preserves the signal information. During quantization, the signal amplitudes changed into step-like levels, showing how continuous values are converted into discrete ones with a small quantization error. The encoded binary output represented the final digital data obtained from the analog signal. The simulation also showed that increasing quantization levels improves signal accuracy but requires a higher bit rate and bandwidth. Overall, the simulation results matched the theoretical expectations and successfully met the objective of understanding analog-to-digital conversion.

Conclusion:

In this experiment, we successfully implemented and analyzed the process of Analog-to-Digital Conversion (ADC) using MATLAB. We explored key concepts such as sampling, quantization, and encoding, which are essential for digital communication systems. The results demonstrated the importance of selecting an appropriate sampling rate and quantization levels to achieve a balance between signal accuracy and transmission efficiency. Understanding these concepts is fundamental for designing reliable digital communication and signal processing systems.

References:

- [1] MathWorks, *MATLAB Documentation*, Natick, MA, USA: MathWorks. [Online]. Available: <https://www.mathworks.com/help/>
- [2] Stormy Attaway, *MATLAB: A Practical Introduction to Programming and Problem Solving*, 6th ed. Boston, MA, USA: Butterworth-Heinemann, 2019.